BMC Women's Health

RESEARCH Open Access

# Maternal mental health improvement in Ghana: role of green spaces exposure

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## Abstract

**Background** Green space exposure has been demonstrated to influence human physical and psychosocial health. We explored the relationship between self-reported green space exposure and maternal perceived stress, anxiety and depression.

**Methods** We interviewed 420 mothers using a hospital-based cross-sectional study design. We assessed maternal stress, anxiety and depression levels. Maternal self-reported green space exposure characteristics were identified using an interviewer-administered questionnaire. Multiple regression was used to identify the nature of relationships.

**Results** Higher perceived stress, anxiety, and depression were observed in 58.1%, 42.5%, and 58.9% of respondents, respectively. Perceived stress was significantly associated with maternal utilization of green spaces for leisure activities and physical exercise (β = -0.118, 95% CI [-0.118, 0.025]). Mothers' belief that spending time in green spaces positively impacts one's overall well-being was significantly associated with maternal anxiety (β = 0.117, 95% CI [0.049, 0.491]) and depression (β = 0.164, 95% CI [0.205, 0.769]).

**Conclusions** Our study findings suggest that green space exposure has the potential to attenuate stress, anxiety and depression among mothers. Policies relating to the establishment of green spaces and the maternal utilization of green spaces should be encouraged. Further studies should explore maternal health green space relationships using objective green space measurement.

**Keywords** Green spaces, Stress, Anxiety, Depression, Maternal health, Sunyani, Ghana

## Introduction

The interplay between self-reported exposure to green spaces and the experience of stressful life events among mothers represents a critical area of inquiry. Rapid urbanization in Sub-Saharan Africa in general, has led to significant alterations in land use, contributing to the deterioration of natural environments and the reduction of accessible green spaces [1, 2]. Green spaces, which include parks, gardens, and other areas with vegetation, have been documented to provide numerous physical and psychological health benefits. These benefits may be especially pertinent for mothers, who often face unique stressors associated with caregiving, social expectations, and socioeconomic challenges [3, 4].

Existing literature has established a correlation between green space exposure and improved mental health outcomes; however, these studies often generalize findings without considering the contextual and demographic variations that influence these benefits. Notably, prior research [5, 6] does not delineate the specific conditions under which green space exposure is most effective

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in mitigating stress and anxiety, particularly among vulnerable populations such as pregnant and lactating mothers. This omission presents a significant research gap, as maternal health is influenced by a complex interplay of socioeconomic stressors, environmental exposures, and psychological resilience mechanisms. Given that maternal stress, anxiety, and depression are critical determinants of neonatal health and developmental outcomes [7, 8]. Investigating the role of green spaces within this specific context is imperative. This study advances current knowledge by exploring how self-reported green space exposure interacts with maternal stressors in a setting characterized by economic constraints, limited social support, and heightened susceptibility to mental health challenges.

Well-documented pregnancy and maternal health aspects often discuss maternal stress and birth outcomes [7, 9, 10], the mental well-being of the offspring [11], neighborhood conditions [12] dietary supplementation [13], fetal growth [14, 15], antenatal care [16–18] delivery complications and risk factors [19, 20], neonatal health [21, 22], and maternal mental health [23–25]. In addition, there have been discussions around the perinatal period (the period from conception to the end of the first postpartum year), which is often characterized by elevated levels of stress, anxiety and depression. These conditions are critical determinants of maternal mental health and neonatal health, yet natural environmental settings that could manage the conditions have not been widely studied.

Biomedically, stress is defined as an organism's response to adverse stimuli. Psychologically, stress relates to the process by which a person and the environment interact [26]. It is often associated with an imbalance between the demands of life and an individual's adaptive capacity [27]. Also, the WHO defines stress as a state of worry or mental tension caused by a difficult situation [28]. Stress accounts for 17% of deaths worldwide and 25% among mothers aged 18 to 35 years [29–31]. High amounts of stress are associated with poor health and mental health for both mothers and their children [29, 32]. Several studies have identified stressful life events as a psychosocial risk factor that influences pregnancy outcomes if not appropriately managed [27, 33, 34]. And such stressful events produce anxiety, especially among expectant mothers.

Stress, a response to adverse stimuli, can manifest psychologically and physiologically, influencing maternal well-being. Chronic stress during pregnancy has been linked to negative maternal and neonatal health outcomes, including increased risks of preterm birth, low birth weight, and developmental impairments [7, 34]. One key manifestation of chronic stress is anxiety, a heightened state of apprehension that can exacerbate maternal distress and contribute to adverse pregnancy outcomes [24].

[35] refers to anxiety as a fearful apprehension that is out of proportion to external circumstances and is accompanied by autonomic hyperactivity symptoms such as palpitations and sweating. Anxiety can lead to obstetric complications during pregnancy [36] and can have serious effects on the offspring's health [37]. For example, anxiety could precipitate shorter gestation and negative neurodevelopment for the infant [8] Risk factors of anxiety in pregnant women include complications in previous pregnancies, history of miscarriage or preterm birth, maternal desire regarding pregnancy, number of abortions, number of cigarettes smoked daily and drug use [38]. In pregnancy, anxiety has been linked with preterm labor [8] low birth weight [39], gestational hypertension [40], and postpartum depression [41].

Depression, a common mental disorder, is characterized by persistent sadness, loss of interest, and impaired daily functioning [31]. During pregnancy and the postpartum period, hormonal changes can contribute to depressive symptoms, which may negatively impact maternal well-being and infant development [42]. Risk factors include single motherhood, low socioeconomic status, and adverse obstetric history [43, 44]. Additionally, high maternal stress during the perinatal period has been linked to adverse birth outcomes, such as preterm birth and developmental delays [8, 45].

Several studies suggest that green spaces contribute to mental well-being and physical health by offering restorative environments that alleviate stress and anxiety [46–50]. Features such as dense tree canopies, lush vegetation, water elements, open grassy areas, and walking trails create a serene and calming atmosphere, fostering relaxation and psychological restoration [51]. Additionally, shaded seating areas, natural soundscapes, and biodiversity enhance the stress-reducing benefits of green spaces by promoting mindfulness and social interaction [52]. Systematic reviews further support the positive association between green space exposure and reduced mental distress, emphasizing the need to integrate such features into urban planning for improved public health [53–55].

Several studies highlight the positive impact of residential greenness on maternal and neonatal health. Research from Australia, China, and Spain consistently indicates that increased exposure to green spaces is linked to better birth outcomes. Higher residential greenness has been associated with increased birth weight [56], reduced risk of adverse pregnancy outcomes [55], and improved neonatal indicators such as birth weight and head circumference [57]. These findings reinforce the significance of green spaces in promoting maternal and infant health,

emphasizing the need for urban planning strategies that integrate natural environments.

Consequently, understanding the role of green spaces as potential buffers against these stressors is essential for developing effective public health interventions and urban planning strategies aimed at improving maternal and child health outcomes in middle-income countries in general and Ghana in particular [58, 59]. There remains a dearth of research focusing on self-reported green space characteristics and stressful life events of mothers in the Sub-Saharan Africa context. This gap is significant, as context-specific data are crucial for informing policy decisions, land use planning, and resource allocation. Therefore, this study aims to contribute to the growing body of literature that explores green space and mental health relationships. Therefore, this study aims to investigate the relationship between self-reported green space exposure and maternal psychological outcomes specifically, stress, anxiety, and depression—among mothers in a hospital-based setting in Sunyani, Ghana. We hypothesize that higher self-reported green space exposure is associated with lower levels of stress, anxiety, and depression. In doing so, we seek to contribute empirical evidence to the discourse on how environmental features can serve as a buffer against psychosocial distress in low- and middle-income urban contexts. To our knowledge, this study represents the first attempt to explore links between self-reported green space exposure and maternal mental health symptoms. By investigating this relationship, the research seeks to provide new insights that can inform community health initiatives, enhance urban planning efforts, and promote equitable access to green spaces.

## Methods

### Study area

The Sunyani Municipality is geographically located between latitudes 7°20'N and 7°5'N and longitudes 2°30'W and 2°10'W [60]. It shares its northern boundary with the Sunyani West Municipality, the western boundary with the Dormaa East District, the southern boundary with the Asutifi District, and the eastern boundary with the Tano North District. The total land area of the municipality encompasses 829.3 square kilometers (320.1 square miles). Sunyani, as the regional capital of the Bono Region, is currently undergoing significant growth in both its spatial dimensions and economic activities [60]. The region has appreciable green spaces in its environments and provides a serene atmosphere for its inhabitants (Fig. 1).

This study was grounded in environmental psychology and stress reduction theories, which posit that natural environments, such as green spaces, can alleviate stress and improve psychological well-being [51, 52]. The Stress Reduction Theory (SRT) by [61] suggests that exposure to natural environments facilitates physiological and psychological recovery from stress, while the Attention Restoration Theory (ART) by [52] highlights the cognitive benefits of natural settings in reducing mental fatigue. These theoretical perspectives guide the study's assumption that green space exposure may buffer the negative effects of stress, anxiety, and depression among mothers [62, 63].

Based on this framework, the study hypothesizes that maternal exposure to green spaces (e.g., parks, natural vegetation, gardens) is associated with lower perceived stress, anxiety, and depression levels. The conceptual model assumes that green space exposure influences mental health outcomes through mechanisms such as psychological restoration, stress buffering, and social interaction opportunities [64, 65].

### Sampling

We sampled 420 mothers who visited the post-natal clinic at the Sunyani Teaching Hospital for child weighing and routine postnatal care. This number represented approximately 70% of the monthly clinic attendees. This recruitment strategy aligns with prior maternal health research that has employed similar methodologies [66–68]. In order to be eligible for the study, a mother's child should have been aged 0 to 24 months. Assistants and caregivers were excluded from the study. Each consenting mother participated in a single structured interview conducted within the hospital setting [69, 70].

The sampling method excluded mothers who do not routinely access institutional postnatal care, such as those in remote or low-income communities with limited healthcare access. This clinic-based recruitment strategy could result in selection bias, as mothers attending postnatal clinics may differ systematically in terms of health awareness, socioeconomic status, or environmental exposure compared to the broader maternal population in the municipality [68, 71]. Therefore, findings may not be generalizable to all mothers in Sunyani, particularly those residing in peri-urban or rural areas.

### Questionnaire design and administration

An interviewer-administered questionnaire was designed to evaluate data collection among mothers. It included questions related to the quantity and quality of greenery, visibility of natural vegetation from residences, accessibility to parks, frequency of visits to natural areas, and the use of green spaces for recreational or physical activities. All questions were closed-ended, allowing participants to select from predefined options. The assessments were adapted from the work of Reid, C. E., Rieves, E. S., and Carlson, K. (2022), which linked these perceptions to improved mental health outcomes among Denver

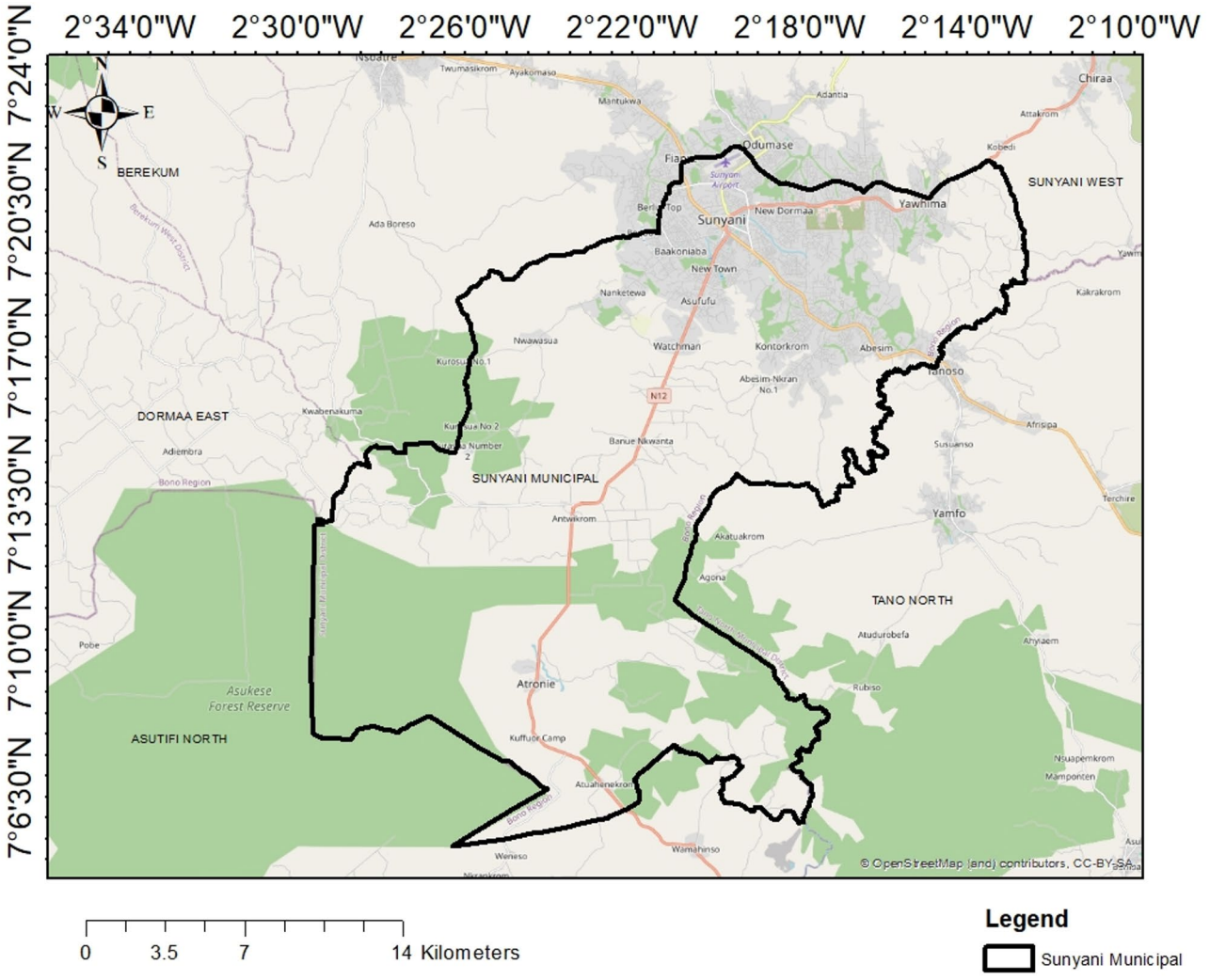



**Fig. 1** Map of the study area

residents during the COVID-19 pandemic, underscoring the significance of green space exposure for mental health and well-being [72]. Additionally, the questionnaire examined the impact of stressful life events on mothers over the past two years. Participants discussed how these events influenced their caregiving abilities and rated the level of stress they experienced. This approach aligned with existing research that highlights the substantial impacts of maternal stress on infant development, emphasizing the need to explore these variables concerning maternal and child health [73, 74].

Four nurses were recruited and trained before the administration of the survey. This was done following recommendations for utilizing qualified and skilled personnel to enhance the ethical rigor and reliability of data collection [75, 76]. The nurses were responsible for explaining the study's objectives to participants, obtaining informed consent, and entering the responses into the questionnaire. Respondents' answers were captured with the use of Android mobile phones. Kobo-collect, a free data collection software, was deployed onto four mobile phones dedicated to data collection. This facilitated the deployment of copies of the questionnaire to each mobile phone and also facilitated synchronous data collection and data entry. Data collection was done from June to August 2024.

### Variables

We defined 3 continuous dependent variables for our study. They were stress, anxiety, and depressed. Eleven categorical maternal socio-demographic characteristics were assessed. These were: maternal age, district of residence, marital status, educational level, employment status and maternal ethnicity. The others were home ownership, parity, monthly household income, household size and frequency of visits to green spaces during pregnancy. Eight self-reported green space characteristics were investigated. All were designated as categorical

variables. These were; How would you describe the amount of greenery or vegetation in your neighbourhood?; coded as '1 = abundant', '2 = moderate', '3 = none' and '4 = scarce'. 'How visible are green spaces or natural vegetation from your residence?', coded as '1 = highly visible', '2 = not very visible', '3 = not visible at all' and '4 = somewhat visible'. 'How easy is it for you to access the closest park or green space?', coded as '1 = difficult', '2 = not very easy', '3 = somewhat easy' and '4 = very easy'. How often do you spend time in areas with natural vegetation? Coded as '1 = daily', '2 = never', '3 = once a week', '4 = rarely', and '5 = several times a week'. How would you rate the quality of green spaces close to your home? Coded as '1 = excellent', '2 = fair', '3 = good' and '4 = poor'. Do you utilize green spaces for leisure activities or physical exercise? Coded as '1 = no never', '2 = no rarely', '3 = yes occasionally', '4 = yes, regularly'. How do you feel after spending time in green spaces? coded as '1 = energized', '2 = no noticeable difference', '3 = refreshed', '4 = relaxed' and '5 = stressed'. Do you believe that spending time in green spaces positively impacts your overall well-being? Coded as '1 = agree', '2 = disagree', '3 = neutral', and '4 = strongly agree' and '5 = strongly disagree'. In addition to maternal socio-demographic characteristics, the study acknowledges that unmeasured confounders such as social support networks, exposure to urban stressors, and neighborhood safety may influence both green space utilization and maternal mental health outcomes. Prior studies underscore the role of perceived safety, noise pollution, and overcrowding as urban stressors that mediate psychological well-being [77, 78]. Similarly, social support from family, partners, and community institutions has been shown to moderate stress and buffer against anxiety and depressive symptoms among mothers [79, 80]. Although these variables were not directly measured in this study, their potential confounding effects are recognized.

### Estimation of green space exposure

Green space exposure can be assessed by self-reported means or by objective means. Self-reported exposure is typically assessed through various survey questions. These include factors such as the abundance of surrounding greenery [81], the quality of green spaces [62, 82], perceptions of safety within green spaces [83, 84], proximity to residential areas [85], and the frequency and duration of use [86]. Some studies further posit that perceptions of green space exposure may have a greater impact on specific health pathways than objective measures [62]. We operationally defined green spaces as areas covered with vegetation, such as parks, gardens, forests, natural reserves, botanical gardens, and open spaces for recreational activities. This study used self-reported or perceived green space exposure to assess its association with maternal stress, anxiety, and depression. Although our study employed self-reported measures, integrating objective indicators like the Normalized Difference Vegetation Index (NDVI), land-use data, or GIS mapping could provide more accurate estimations of green space access and quality. However, these tools require precise geolocation data, which was not feasible in the hospital-based setting due to ethical concerns and mobility among participants.

While self-reported green space exposure allows for capturing subjective experiences, it is also susceptible to recall bias and perceptual inconsistencies, which may compromise data validity. Perceptions of greenness can differ based on individual mental states, environmental awareness, or social context [87]. Unlike objective measures such as the Normalized Difference Vegetation Index (NDVI) or Geographic Information Systems (GIS), which quantify vegetation cover and proximity using satellite imagery and spatial analysis [88], subjective assessments may reflect personal or cultural biases rather than actual environmental exposure. This discrepancy could potentially attenuate or inflate associations between green space and psychological outcomes.

By evaluating mothers' perceptions of greenery abundance, accessibility, and usage patterns, the study aimed to determine how exposure to green spaces influences maternal mental health outcomes.

### Maternal stressful life events

#### *Estimation of stress*

The Perceived Stress Scale (PSS-9) used in this study is a shortened adaptation of the original 14-item scale developed by Cohen and colleagues [89]. This scale assesses respondents' stress levels based on their answers to their perceived control over life circumstances, confidence in handling personal, belief that life is proceeding as expected, and feelings of being overwhelmed. In our Perceived Stress Scale (PSS-9), respondents answered 9 Likert scale questions. All answers in this scale were coded and scored as follows; '0 = Never', '1 = almost never', '2 = sometimes', '3 = fairly often', and '4 = very often' (Table 1). We summed the average responses to each question and divided them by the total number of questions. The aggregated score of 16.52 yielded a weighted average of approximately 1.84. A mean score of 1.84 or above indicated a heightened perception of stress associated with specific items, underscoring their significant influence on the respondents. Conversely, a score below 1.84 denoted a lower level of stress perception, suggesting that respondents experienced reduced concern regarding those stressors. This methodology allowed for the categorization of each item as either above or below the established stress perception threshold of 1.84. We classified responses above the threshold as moderate levels

**Table 1** Stress scale of mothers

| Question | Never | Almost never | Sometimes | Fairly often | Very often | Mean | Sd | Decisions |
|---|---|---|---|---|---|---|---|---|
| In the last month, how often have you been upset because of something that happened unexpectedly? | 31 (7.5%) | 121 (29.2%) | 163 (39.4%) | 81 (19.6%) | 18 (4.3%) | 1.84 | 0.969 | The decision threshold met (around the average) |
| In the last month, how often have you felt that you were unable to control the important things in your life? | 62 (15.0%) | 129 (31.2%) | 138 (33.3%) | 73 (17.6%) | 12 (2.9%) | 1.62 | 1.031 | Lower perceived stress (below average) |
| In the last month, how often have you felt nervous and 'stressed'? | 29 (7.0%) | 113 (27.1%) | 149 (35.7%) | 98 (23.5%) | 28 (6.7%) | 1.96 | 1.026 | Higher perceived stress (above average) |
| In the last month, how often have you felt confident about your ability to handle your personal problems? | 33 (7.9%) | 99 (23.7%) | 115 (27.5%) | 129 (30.9%) | 42 (10.0%) | 2.11 | 1.119 | Higher perceived stress (above average) |
| In the last month, how often have you felt that things were going your way? | 33 (7.9%) | 138 (33.1%) | 163 (39.1%) | 76 (18.2%) | 7 (1.7%) | 1.73 | 0.908 | Lower perceived stress (below average) |
| In the last month, how often have you found that you could not cope with all the things that you had to do? | 40 (9.7%) | 142 (34.4%) | 133 (32.2%) | 87 (21.1%) | 11 (2.7%) | 1.73 | 0.988 | Lower perceived stress (below average) |
| In the last month, how often have you been able to control irritations in your life? | 29 (7.0%) | 109 (26.3%) | 106 (25.5%) | 127 (30.6%) | 44 (10.6%) | 2.12 | 1.123 | Higher perceived stress (above average) |
| In the last month, how often have you been angered because of things that were outside of your control? | 34 (8.2%) | 158 (37.9%) | 142 (34.1%) | 72 (17.3%) | 11 (2.6%) | 1.68 | 0.941 | Lower perceived stress (below average) |
| In the last month, how often have you felt difficulties were piling up so high that you could not overcome them? | 34 (8.2%) | 139 (33.5%) | 162 (39.0%) | 67 (16.1%) | 13 (3.1%) | 1.73 | 0.936 | Lower perceived stress (below average) |

of stress and pointed to the potential need for intervention. Lower scores on items related to mothers' feelings of control over life events pointed to an ability to manage responsibilities and suggested a reduced experience of stress. This distinction allowed for a nuanced interpretation of stress levels, categorizing respondents into moderate or low-stress perception groups and identifying potential intervention needs.

While our threshold of 1.84 was based on the internal sample mean, we recognize that this cut-off corresponds to the 50th percentile of our sample's distribution and aligns with prior studies using PSS-10/PSS-14 that define moderate stress at a comparable.

3-item-level mean of 1.7–2.0 [90, 91]. Although no PSS-9-specific clinical norm exists for Ghana, these thresholds approximate global norms for moderate stress in general adult populations. We acknowledge that future work could validate a Ghana-specific cut-off via normative sampling or clinical interviews.

#### *Estimation of anxiety*

We adopted the Hamilton Anxiety Rating Scale (HAM-A) [92] to assess respondents' anxiety levels. This scale evaluates various anxiety-related symptoms, including nervousness, concentration difficulties, stress, and sleep disturbances [93]. It also assessed the severity of anxiety symptoms, including mood swings, somatic symptoms, and cognitive issues related to anxiety. Respondents answered 14 Likert scale questions, coterminous with symptoms, in relation to the anxiety scale. All answers in this scale were coded and scored as follows: '0 = not present', '1 = mild', '2 = moderate', '3 = severe', and '4 = very severe' (Table 2). By averaging the mean values for each question, it resulted in a total sum of 18.33. A weighted average of 1.31 was derived by dividing this sum by the total number of questions. This established a baseline for the average perceived anxiety. Symptoms that scored above this threshold were classified as higher perceived anxiety, while those scoring below indicated lower perceived anxiety. This threshold approximates the moderate anxiety range used in prior HAM-A validation studies [94, 95]. Although global norms for item-level HAM-A in perinatal settings vary (typically 1.2–1.5), our chosen cut-off reflects our sample's median and is consistent with clinically meaningful anxiety levels.

#### *Estimation of depression*

The 10-item Center for Epidemiological Studies Depression Scale (CES-D-10) was adopted to assess depression for its efficacy in screening for depression [96]. Ours was adapted from [97]. The CES-D-10 measures depressive symptoms such as feelings of sadness, hopelessness, and loneliness. It also evaluated symptoms of depression experienced during the previous week, encompassing

**Table 2** Anxiety scale (HAM-A) of mothers

| Symptom | Not present | Mild | Moderate | Severe | Very severe | Mean | Sd | Decisions |
|---|---|---|---|---|---|---|---|---|
| Worries anticipation of the worst, fearful anticipation, irritability | 32 (7.6%) | 224 (53.3%) | 130 (31.0%) | 27 (6.4%) | 7 (1.7%) | 1.41 | 0.790 | Higher perceived anxiety (above average) |
| Feelings of tension, fatigability, startle response, moved to tears easily, trembling, feelings of restlessness, inability to relax | 93 (22.2%) | 153 (53.3%) | 126 (31.0%) | 42 (6.4%) | 5 (1.7%) | 1.32 | 0.966 | Around average anxiety |
| Fear of dark, strangers, being left alone, animals, traffic, crowds | 142 (34.0%) | 139 (53.3%) | 100 (31.0%) | 34 (6.4%) | 3 (1.7%) | 1.08 | 0.982 | Lower perceived anxiety (below average) |
| Difficulty in falling asleep, broken sleep, unsatisfying sleep, fatigue on waking, dreams, nightmares, night terrors | 65 (15.6%) | 173 (53.3%) | 122 (31.0%) | 49 (6.4%) | 7 (1.7%) | 1.42 | 0.946 | Higher perceived anxiety (above average) |
| Difficulty in concentration, poor memory | 72 (17.3%) | 150 (53.3%) | 139 (31.0%) | 46 (6.4%) | 9 (1.7%) | 1.45 | 0.973 | Higher perceived anxiety (above average) |
| Loss of interest, lack of pleasure in hobbies, depression, early waking, diurnal swing | 83 (19.9%) | 175 (53.3%) | 115 (31.0%) | 39 (6.4%) | 5 (1.7%) | 1.30 | 0.932 | Around average anxiety |
| Pains and aches, unsteady voice, twitching, stiffness, myoclonic jerks, grinding of teeth, increased muscular tone | 75 (17.9%) | 184 (53.3%) | 110 (31.0%) | 42 (6.4%) | 7 (1.7%) | 1.33 | 0.941 | Around average anxiety |
| Blurring of vision, hot and cold flushes, feelings of weakness, pricking sensations | 70 (16.9%) | 165 (53.3%) | 127 (31.0%) | 47 (6.4%) | 5 (1.7%) | 1.40 | 0.938 | Higher perceived anxiety (above average) |
| Pain in the chest, throbbing of vessels, fainting, feelings of missing a beat | 85 (20.5%) | 173 (53.3%) | 2 (1.7%) | 1.29 | 2 (1.7%) | 1.29 | 0.930 | Around average anxiety |
| Pressure or constriction in the chest, choking feelings, sighing, dyspnea | 71 (17.0%) | 183 (53.3%) | 119 (31.0%) | 38 (6.4%) | 7 (1.7%) | 1.35 | 0.922 | Around average anxiety |
| Difficulty in swallowing, wind, abdominal pain, burning sensations, abdominal fullness, nausea, vomiting, borborygmi, looseness of bowels, loss of weight, constipation | 93 (22.5%) | 183 (53.3%) | 96 (31.0%) | 34 (6.4%) | 8 (1.7%) | 1.23 | 0.953 | Around average anxiety |
| Frequency and urgency of micturition, amenorrhea, menorrhagia, development of frigidity, loss of libido | 77 (18.6%) | 184 (53.3%) | 115 (31.0%) | 37 (6.4%) | 1 (1.7%) | 1.28 | 0.876 | Lower perceived anxiety (below average) |
| Dry mouth, flushing, pallor, tendency to sweat, giddiness, tension headache, raising of hair | 84 (20.0%) | 173 (53.3%) | 118 (31.0%) | 41 (6.4%) | 3 (1.7%) | 1.30 | 0.922 | Around average anxiety |
| Fidgeting, restlessness or pacing, tremor of hands, furrowed brow, strained face, sighing or rapid respiration, facial pallor, swallowing, etc. | 77 (18.7%) | 219 (53.3%) | 87 (31.0%) | 26 (6.4%) | 2 (1.7%) | 1.17 | 0.819 | Lower perceived anxiety (below average) |

emotional, physical, and social aspects among mothers. In our depression scale (CES-D-9), respondents answered 9 Likert scale questions. All answers in this scale were coded and scored as follows: '0 = rarely', '1 = little of the time', '2 = occasionally', '3 = most of the time' (Table 3). We employed a weighted technique that involved summing the mean values of the nine symptoms on the depression scale to assess the degree of depression among respondents. By calculating a weighted average of 1.18 (derived from 10.65 divided by 9) for each symptom, the study established a baseline for the level of reported depression. A symptom was classified as indicative of a higher degree of depression when its mean score exceeded 1.18. Conversely, mean scores below this threshold suggested that respondents' experiences with depression were less affected by these symptoms, thereby enhancing the understanding of each symptom's relationship to the group's overall depression and its significance. Although CES-D-10 cut-offs vary, a total score ≥ 10 (mean ≈ 1.0) is commonly used to indicate clinically significant depressive symptoms in community samples [97, 98]. Our sample's mean item threshold of 1.18 corresponds to a total CES-D-10 score of roughly 10.6, consistent with validated norms for moderate depression in diverse adult populations.

### Data analysis

The quantitative data obtained from the questionnaires were analyzed using Jeffreys's Amazing Statistics Program (JASP). The JASP software is a free statistical application that is supported by the University of Amsterdam [99]. Descriptive statistics, such as means, percentages, and standard deviations, were calculated to summarize the data. Perceived Stress Scale (PSS-4) (Table 2), Depression Scale (CES-D) (Table 3), and the Anxiety Scale (HAM-A) (Table 4) among the study population were examined utilizing a weighted average method. This approach has been applied in studies by [89, 97] and [92]. This approach to methodology was deemed essential for providing insights into the psychological well-being of our study population [92, 97, 100]. Multiple regression analysis was used to explore relationships between green space characteristics in the Perceived Stress Scale (PSS-4), Depression Scale (CES-D), and Anxiety Scale (HAM-A) [72].

## Results

### Descriptive characteristics

Table 5 shows a summary descriptive characteristics of the respondents. The majority of the mothers were between the ages of 31 and 35 (48.4%), followed by those aged 26 to 30 (25.8%). More than half of the study population were married (55.2%) with more than 40% employed. Also, most mothers lived in the Sunyani municipality (68.8%) and belonged to the Akan ethnic group (70%). Few mothers owned their own homes (28.3%). Additionally, less than 30% had 3 or more deliveries. Most mothers (74.5%) reported having a monthly income of 1600 Ghana Cedis or less. Fewer than 10% regularly visited green spaces during pregnancy. Most of the study population was observed to have high perceived stress (58.1%), high anxiety (57.5%), and low depression levels (41.1%).

### Perceived stress of mothers

Based on the analysis, all mothers in the study experienced some level of stress, but it was highly perceived by 58.1% of the respondents. This suggests that while some mothers reported moderate or lower stress levels, more than half experienced elevated stress symptoms. Higher perceived stress was particularly observed in mothers who scored above the threshold on key items related to feelings of nervousness, confidence in handling personal

**Table 3** Depression scale (CES-D-9) of mothers. ($N = 420$)

| Symptom | Rarely | Little of the time | Occasionally | Most of the time | Mean | SD | Decisions |
|---|---|---|---|---|---|---|---|
| I was bothered by things that usually don't bother me | 97 (23.3%) | 242 (58.0%) | 71 (17.0%) | 7 (1.7%) | 0.97 | 0.686 | Lower perceived depression (below average) |
| I had trouble keeping my mind on what I was doing | 108 (26.0%) | 190 (58.0%) | 99 (17.0%) | 19 (1.7%) | 1.07 | 0.823 | Higher perceived depression (above average) |
| I felt depressed | 97 (23.3%) | 210 (58.0%) | 96 (17.0%) | 14 (1.7%) | 1.06 | 0.771 | Higher perceived depression (above average) |
| I felt that everything I did was an effort | 114 (27.3%) | 202 (58.0%) | 96 (17.0%) | 6 (1.7%) | 0.99 | 0.749 | Lower perceived depression (below average) |
| I felt hopeful about the future | 77 (18.6%) | 222 (58.0%) | 94 (17.0%) | 22 (1.7%) | 1.15 | 0.777 | Higher perceived depression (above average) |
| I felt fearful | 94 (22.5%) | 203 (58.0%) | 105 (17.0%) | 15 (1.7%) | 1.10 | 0.783 | Higher perceived depression (above average) |
| My sleep was restless | 92 (22.0%) | 220 (58.0%) | 88 (17.0%) | 18 (1.7%) | 1.08 | 0.774 | Higher perceived depression (above average) |
| I was happy | 0 (0.0%) | 219 (53.3%) | 20 (4.9%) | 78 (19.0%) | 2.11 | 1.276 | Much lower perceived depression (well above average happiness) |
| I felt lonely | 81 (19.6%) | 212 (58.0%) | 111 (17.0%) | 10 (1.7%) | 1.12 | 0.740 | Higher perceived depression (above average) |

**Table 4** Multivariate regression of greenspace exposure, stress, anxiety and depression

| | PSS Stress, *N* = 420 | | | Anxiety Scale HAM-A, *N* = 420 | | | Depression Scale (CES-D-10 ) *N* = 420 | | |
|---|---|---|---|---|---|---|---|---|---|
| **Green space exposure** | **Beta** | **95% CI** | **P-Value** | **Beta** | **95% CI** | **P-Value** | **Beta** | **95% CI** | **P-Value** |
| How would you describe the amount of greenery or vegetation in your neighborhood | 0.031 | -1.21, 0.233 | 0.534 | −0.028 | −0.2.43, 0.133 | 0.565 | −0.028 | −0.453, 0.027 | 0.082 |
| How visible are green spaces or natural vegetation from your residence? | −0.034 | −0.226, 0.108 | 0.485 | −0.013 | −0.2.00, 0.154 | 0.796 | −0.002 | −0.232, −0.232 | 0.962 |
| How easy is it for you to access the closest park or green space? | −0.018 | −0.146, 0.100 | 0.712 | 0.042 | −0.076, 0.191 | 0.396 | 0.021 | −0.131, 0.203 | 0.672 |
| How often do you spend time in areas with natural vegetation? | 0.046 | −0.092, 0.260 | 0.350 | −0.037 | −0.261, 0.117 | 0.454 | 0.028 | −0.170, 0.311 | 0.564 |
| How would you rate the quality of green spaces close to your home? | 0.030 | −0.084, 0.159 | 0.542 | −0.067 | −0.220, 0.040 | 0.173 | 0.043 | 0.043, 0.043 | 0.379 |
| Do you utilize green spaces for leisure activities or physical exercise? | −0.118 | −0.118, −0.025 | 0.016 | −0.067 | −0.197, 0.035 | 0.169 | −0.062 | −0.243, 0.052 | 0.203 |
| How do you feel after spending time in green spaces? | 0.038 | −0.110, 0.250 | 0.442 | −0.037 | −0.264, 0.118 | 0.453 | −0.050 | −0.372, 0.118 | 0.309 |
| Do you believe that spending time in green spaces positively impacts your overall well-being? | 0.030 | −0.145, 0.274 | 0.546 | 0.117 | 0.049, 0.491 | 0.017 | 0.164 | 0.205, 0.769 | 0.001 |

challenges, and ability to manage daily irritations. These findings suggest that while some respondents demonstrated effective stress management, others experienced heightened stress levels that may require intervention.

More than two-thirds of mothers had reportedly felt nervous and stressed in the preceding month. A similar proportion of mothers felt confident about their ability to handle their personal problems. Over 30% of mothers were unable to control irritations in their lives in the previous month. The mean of 1.62 suggests that higher perceived stress was observed. Lower perceived stress was observed among mothers who felt that they were able to control the important things in their lives 15% More than 50% of mothers indicated that they felt that things were going their way in the previous month. Fewer than 30% of mothers could not cope with all the things they had to do. Furthermore, only 20% of mothers often felt angered because of things that were outside of their control. Mothers who rarely felt overwhelmed by accumulating difficulties reported lower perceived stress levels.

While respondents reported competence in managing these stressors, higher scores on the personal problem-handling, stress, and irritation management scales indicated elevated stress levels [64, 101]. Overall, the investigation revealed disparities in stress levels across different life domains and emphasized the potential mental health benefits of exposure to natural environments, which may alleviate stress and enhance maternal well-being [62].

#### Perceived anxiety of mothers

In this study, there was evidence of perceived anxiety among mothers 42.5%. Higher perceived anxiety was found in mothers who had symptoms such as worrying, anticipation of the worst, fearful anticipation and irritability 39.1%. Also, mothers who had difficulty in falling asleep, broken sleep, unsatisfying sleep, fatigue on waking, dreams, nightmares, night terrors had higher perceived anxiety. We found difficulty in concentration and poor memory in mothers who had high perceived anxiety. In addition, the blurring of vision, hot and cold flushes, feelings of weakness, pricking sensations were observed in mothers with higher perceived anxiety.

Average anxiety was observed among mothers who expressed feelings of tension, fatigability, startle response, moved to tears easily, trembling, feelings of restlessness, inability to relax. Mothers who had loss of interest, lack of pleasure in hobbies, depression, early waking, diurnal swing, Pains and aches, unsteady voice, twitching, stiffness, myoclonic jerks, grinding of teeth and increased muscular tone were observed to have average anxiety.

Also, pain in the chest, throbbing of vessels, fainting, feelings of missing a beat, pressure or constriction in the chest, choking feelings, sighing, dyspnea were symptoms observed in mothers with average anxiety.

Mothers with symptoms such as difficulty in swallowing, wind, abdominal pain, burning sensations, abdominal fullness, nausea, vomiting, borborygmi, looseness of bowels, loss of weight and constipation were also observed to have average anxiety.

Dry mouth, flushing, pallor, tendency to sweat, giddiness, tension headache, and raising of hair were also reported by mothers with average anxiety.

Mothers who reported symptoms such as fear of the dark, strangers, being left alone, animals, traffic and crowds had lower perceived anxiety. Also, frequency and urgency of micturition, amenorrhea, menorrhagia, development of frigidity, and loss of libido were observed in mothers with less anxiety. In addition, less anxiety was observed in mothers who reported having symptoms

**Table 5** Socio-demographic characteristics of respondents

| Variable | Frequency | Percentage |
|---|---|---|
| **Maternal Age** | | |
| under 20 | 8 | 1.9% |
| 21 to 25 | 22 | 5.2% |
| 26 to 30 | 108 | 25.8% |
| 31 to 35 | 203 | 48.4% |
| 36 and older | 78 | 18.6 |
| District of Residence | | |
| Municipal | 289 | 68.8% |
| West | 130 | 31.2% |
| Marital Status | | |
| Married | 232 | 55.2% |
| Not married | 88 | 44.8% |
| Educational Level | | |
| No formal education | 11 | 2.6% |
| Primary education | 54 | 12.9% |
| Secondary education | 161 | 38.3% |
| Tertiary education | 194 | 46.2% |
| Employment Status | | |
| Employed | 273 | 65.3% |
| Unemployed | 145 | 34.7% |
| Maternal Ethnicity | | |
| Akan | 292 | 70.0% |
| Ewe | 37 | 8.9% |
| Ga-Adangbe | 5 | 1.2% |
| Hausa | 50 | 12.0% |
| Mole Dagbani | 33 | 7.9% |
| Home ownership | | |
| Owner | 119 | 28.3% |
| Rental | 301 | 71.7% |
| Parity | | |
| ≤ 3 | 297 | 70.7% |
| Above 3 | 123 | 29.3% |
| Monthly household income | | |
| 1600 Ghana Cedis or less | 313 | 74.52 |
| More than 1600 Ghana Cedis | 107 | 25.48 |
| Household size | | |
| 3 and below | 94 | 22.43% |
| 4 and above | 325 | 77.57% |
| How often did you visit green spaces during pregnancy? | | |
| Daily | 38 | 10.0% |
| Monthly | 64 | 16.8% |
| Never | 3 | 0.8% |
| Rarely | 74 | 19.4% |
| Weekly | 202 | 53.0% |
| Stress | | |
| High | 244 | 58.1% |
| Low | 176 | 41.9% |
| Anxiety | | |
| High | 178 | 42.5% |
| Low | 242 | 57.5% |
| Depression | | |
| High | 247 | 58.9% |
| Low | 173 | 41.1% |

such as fidgeting, restlessness or pacing, tremor of hands, furrowed brow, strained face, sighing or rapid respiration and facial pallor.

#### Perceived depression of mothers

Obtained results from the study provided evidence of perceived depression among the study population. Depressive symptoms were present in the study population. Higher perceived depression was observed in mothers who reported that [1], I had trouble keeping my mind on what I was doing [2],, I felt depressed [3],, I felt hopeful about the future [4],, I felt fearful [5],, My sleep was restless, and [6], I felt lonely.

Lower perceived depression was observed in mothers who reported that they were not bothered by things that usually don't bother them. Few mothers (1.7%) indicated that they felt that everything they did was an effort. However, lower perceived depression was observed in respondents who answered this question. Mothers (19.0%) who indicated that they were happy, had lower perceived depression.

#### Multivariate regression of stress, anxiety, depression and green space exposure among mothers

We used a multivariate model to assess the association between exposure to green spaces and three dependent variables; stress, anxiety, and depression (Table 4). This analysis utilized various parameters of green space exposure, including visibility, accessibility, quantity, frequency of use, quality, leisure use, emotional effects, and personal perceptions regarding its benefits, to evaluate their impact on psychological well-being.

The findings revealed that most relationships between psychological outcomes and exposure to green spaces were statistically insignificant, suggesting that mere proximity to or access to green spaces did not significantly affect the psychological health of the mothers. However, notable exceptions were identified. Specifically, engagement in exercise or leisure activities within green spaces was associated with a strong negative correlation with stress levels ($\beta$ = -0.118, 95% CI [-0.118, 0.025]), suggesting that participation in leisure activities in these environments may reduce stress among mothers. In contrast, there was a significant positive relationship between depressive symptoms and beliefs in the beneficial effects of green spaces ($\beta = 0.164$, 95% CI [0.205, 0.769]), as well as a positive correlation between anxiety and those beliefs ($\beta = 0.117$, 95% CI [0.049, 0.491]). While this may seem paradoxical, several plausible explanations emerge. First, individuals with greater psychological distress may be more inclined to recognize or idealize the restorative potential of green spaces, not because they benefit from it directly, but because they perceive it as a desirable yet inaccessible coping resource, a phenomenon described

in environmental psychology as the 'unmet need effect' [102, 103]. Second, belief in the value of green space may reflect a heightened awareness of its absence or poor quality in the local environment, especially in low-resource urban settings, thereby reinforcing dissatisfaction and distress [104]. Third, it is also possible that internalized stigma around emotional vulnerability may lead mothers to express environmental preferences (for example, "green space would help me") as an indirect or socially acceptable disclosure of emotional need, rather than openly discussing mental health symptoms.

This implies that an individual's perceptions of the advantages of green spaces may exert a more substantial influence on psychological outcomes than the actual use or quality of these environments.

These results suggest that green spaces play a crucial role in the psychological benefits derived from these areas than the mere physical presence of the spaces themselves. The unexpectedly strong positive association between elevated anxiety scores and belief in the benefits of green spaces may indicate the presence of confounding variables influencing these relationships, such as societal, environmental, or personal challenges.

While most green space variables were statistically insignificant, the significant positive association between belief in the benefits of green spaces and elevated anxiety and depression was counterintuitive.

Several methodological and contextual factors may explain these findings. First, exposure misclassification may have occurred due to the exclusive use of self-reported green space variables, which are prone to recall and perception biases, especially in subjective dimensions such as quality and accessibility [87]. Second, the cross-sectional design may have obscured temporal effects or delayed psychological benefits, as the mental health impact of nature exposure may require sustained interaction over time to manifest [65, 105]. Third, underlying socioeconomic stressors, including poverty, household overcrowding, and limited healthcare access, may have had a more dominant effect on maternal mental health, potentially masking the more modest effects of green space exposure [63, 106]. Lastly, a form of 'attitude-behavior incongruence,' where mothers facing psychological distress develop stronger beliefs in the value of green spaces as a compensatory coping mechanism, but are unable to access or utilize them due to contextual barriers [106, 107]. Such findings may also reflect the 'availability-expectancy paradox,' whereby positive expectations about green spaces heighten awareness of their absence or inaccessibility, resulting in frustration or distress [108].

## Discussions

We assessed maternal stress, anxiety, and depression and their relationship with self-reported green space exposure characteristics. In terms of perceived stress, mothers report high (30.9%) or very high 58.1% levels of stress, which can impair the cognitive and emotional processes involved in nurturing and parenting [109, 110]. This elevated level of perceived stress possibly reflects mothers' challenges in managing personal hardships while ensuring adequate childcare. The effects of stressful events on mothers' well-being and their ability to care for their children varied; 34.9% of mothers reported a noticeable decline in their well-being and caregiving ability, while 36.1% indicated no significant impact. These findings align with the research of [111] who demonstrated that stress significantly affects parenting efficacy by impacting a mother's ability to manage caregiving responsibilities. These findings suggest that while some mothers struggle with stress-related declines in caregiving ability, others demonstrate resilience, potentially benefiting from social support networks and coping mechanisms. This highlights the need for targeted interventions to support vulnerable mothers while strengthening community-based resources that enhance maternal well-being. This underscores the importance of social and psychological resources in mitigating the negative consequences of stress [112].

Symptoms that significantly impacted daily functioning, such as worries, trouble sleeping, and cognitive impairments, exhibited mean scores exceeding the threshold (1.41, 1.42, and 1.45, respectively) from Table 2. These symptoms could be notably alleviated by the calming effects of green spaces, which have been shown to enhance sleep quality and cognitive performance [51, 65, 113]. Conversely, symptoms such as gastrointestinal issues, fear of external influences, and bodily restlessness scored below the threshold, suggesting lower levels of anxiety that were less distressing for respondents. Exposure to natural environments has also been demonstrated to mitigate these symptoms, contributing to a reduction in physiological manifestations of anxiety and fostering a sense of security [61, 62].

A national study on Canadian women also concluded that residential proximity to vegetation contributed to healthier pregnancies [114]. According to [51], green spaces essentially work to mitigate stress, anxiety, and depression by restoring mental fatigue and enhancing emotional well-being mechanisms.

Research has shown that access to green spaces is often inequitable, with marginalized communities facing greater barriers to enjoying these environments [115]. This inequity can exacerbate the stress experienced by mothers, further highlighting the need for a nuanced

understanding of how environmental factors intersect with maternal health outcomes.

Moreover, average scores for symptoms such as aches, cardiac problems, and respiratory complaints (1.33, 1.29, and 1.35) indicated a mild association with anxiety, suggesting that green spaces may help alleviate the physiological effects of stress [113, 116]. The analysis of the Depression Scale, utilizing a weighted average threshold of 1.18, revealed that the majority of symptoms reported by respondents scored below this threshold, indicating a reduced level of depression. These findings suggest that stress levels were not significantly elevated and may be alleviated by exposure to green spaces, which are known for their calming effects and positive influence on focus and mood. Specifically, symptoms such as reduced irritation, difficulties with concentration, feelings of sadness, and challenges in daily activities all registered below the threshold [62, 63, 113]. Furthermore, happiness scores were notably higher than the cutoff, reflecting low levels of depression and overall positive mental health. This affirms earlier research that reported that green spaces can enhance mood and alleviate symptoms of depression [116]. These natural environments not only foster relaxation and social interaction but also contribute to reduced feelings of loneliness and improved sleep quality [65, 113]. Collectively, the results underscore the potentially significant impact of access to green spaces on reducing depression and enhancing mental health.

In the multivariate analysis, we observed that the utilization of green spaces for leisure activities and physical exercise was negatively associated with stress. This finding is consistent with previous research by [51], who found that physical exercise in natural environments was associated with reduced stress and improved mood. This finding could be explained in part by the restorative effects of green spaces, which provide opportunities for relaxation, reduce mental fatigue, and promote overall well-being [52]. The belief that spending time in green spaces positively impacts overall well-being was significantly associated with maternal anxiety and depression. Similar findings were noted by [63], in their study on the effects of green spaces on mental health. They noted that individuals who perceive green spaces as beneficial for their health tend to experience lower levels of anxiety and depression, highlighting the importance of subjective perceptions of green space quality in fostering mental health. We did not find any significant relationships between perceived stress, anxiety and depression with several green space factors. The presence, accessibility, visibility, and quality of green spaces significantly impact mental and physical well-being. Research indicates that even passive exposure to greenery can reduce stress and enhance mood [51, 65], while ease of access encourages physical activity and psychological benefits [52, 63]. Frequent visits to well-maintained green spaces improve cognitive function and emotional resilience [51, 62], reinforcing their role in stress reduction. High-quality parks with diverse vegetation and amenities further promote relaxation and positive emotional responses [55, 114]. Supported by Ulrich's Stress Reduction Theory and Kaplan's Attention Restoration Theory, these findings highlight the need for urban planning that prioritizes accessible and well-maintained green environments to enhance public health and well-being.

Our research has some limitations. Our study was cross-sectional and did not afford an investigation into causal relationships. Mothers were asked to recall their exposure to green spaces and were also asked to provide self-reported answers on stress, anxiety, and depression. The need for recall may have introduced recall bias. The cross-sectional design also restricted the ability to draw causal inferences between green space exposure and maternal mental health outcomes. While the associations observed suggest potential pathways, the directionality of these relationships cannot be confirmed. For instance, it is plausible that mothers with lower stress or better mental health are more inclined to spend time in green spaces, rather than green spaces causing improvements in well-being, raising the issue of reverse causation [105, 117]. Similarly, mothers experiencing high levels of stress or depression may be less motivated to seek restorative environments, which could understate the beneficial effects of green spaces. These concerns around temporality are consistent with other literature in environmental epidemiology and stress research, where longitudinal designs are required to establish directionality [63, 118].

Objective measures of greenspace exposure at the home or neighbourhoood level could have afforded additional insights into exposure. Previous research suggests that objective measures of green space exposure, such as satellite data or land-use/land-cover data, including the Normalized Difference Vegetation Index (NDVI) and tree canopy coverage, do not always align with subjective perceptions [119–121]. Due to the hospital-based nature of the study, we could not follow up with mothers to find out their actual residential location for validation.

The interpretation of green space benefits was situated within Ghana's sociocultural and gender dynamics. In many Ghanaian households, women bear the bulk of unpaid caregiving and household labor, which reduces their discretionary time for leisure or restorative outdoor activities [122]. Furthermore, cultural stigmas surrounding mental health may influence self-reporting and underdiagnosis, thereby obscuring the true mental health burden [123]. Traditional gender roles, wherein women often bear the brunt of household and caregiving responsibilities, can amplify psychological burdens, particularly in the postpartum period [124, 125]. These gendered

expectations may limit mothers' time and opportunity to engage with green spaces, thereby attenuating their potential benefits.

Urban inequalities also limit access to quality green spaces, especially for low-income households residing in informal settlements [126]. These structural issues may restrict the pathways through which green spaces affect maternal mental health, reinforcing the need for equity-based urban planning. Our exclusive use of self-reporting for mental health outcomes may have introduced social desirability bias, particularly in a setting where mental illness is stigmatized and seldom openly discussed [127]. This might have led to underreporting of symptoms, especially among more educated mothers aware of social norms. Additionally, the hospital-based sampling approach may introduce selection bias, as mothers who attend postnatal clinics at a tertiary facility may differ systematically from those who do not—potentially in socioeconomic status, health-seeking behavior, or environmental exposure. This limits the generalizability of the findings to the broader maternal population, especially those in peri-urban or rural areas. Second, although the models adjusted for several socio-demographic factors, we were unable to control for key potential confounders such as levels of physical activity, perceived social support, or presence of chronic illness, which are independently associated with both green space use and mental health outcomes [63, 128]. The absence of these variables may result in residual confounding. Third, as previously noted, the cross-sectional nature of the study prevents any inference about causality or temporal direction between exposure and outcomes. Despite these limitations, our study points to evidence that suggests that green space exposure characteristics may influence maternal stress, anxiety, and depression. Our study adds to the growing body of literature on greenspace-maternal health relationships.

## Conclusions

The present study assessed maternal stress, anxiety, and depression and their association with self-reported green space exposure characteristics. Stress was associated with the utilization of green spaces for leisure activities and physical exercise. Anxiety and depression were associated with the belief that spending time in green spaces positively impacted overall well-being. More than a third of respondents showed stress, anxiety, and depressive symptoms. This suggests that there may be a need for health care planning policy and research to be tuned towards the incorporation of greenspace exposure to address maternal mental health. These findings underscore the importance of urban planning and public health initiatives aimed at improving access to green spaces. By raising public awareness and facilitating access to these natural resources through urban planning and community engagement, anxiety symptoms can be significantly reduced, thereby enhancing overall community well-being [63, 129]. Future research could use longitudinal approaches to examining green space and maternal–child health relationships.

### Abbreviations

| | |
|---|---|
| CES-D | Center for Epidemiological Studies Depression Scale |
| HAM-A | Hamilton Anxiety Rating Scale |
| JASP | Jeffreys's Amazing Statistics Program |
| PSS-4 | Perceived Stress Scale (4–item version) |
| GSS | Ghana Statistical Service |
| WHO | World Health Organization |
| SD | Standard Deviation |
| CI | Confidence Interval |
| NDVI | Normalized Difference Vegetation Index |
| PHC | Population and Housing Census |
| NICU | Neonatal Intensive Care Unit |

### Acknowledgements

The authors would like to thank the Sunyani Teaching Hospital for granting access to the study participants.

### Author contributions

LD and SKB conceptualized and designed the study. LD and SKB developed the methodology and were responsible for data curation. SKB, LD, and MA conducted the formal analysis and investigation. SKB, LD, and MA contributed to the supervision, validation, and visualization of the study. SKB, LD, and MA drafted the original manuscript. LD, MA, and SKB critically reviewed, revised, and finalized the manuscript. All authors have read and approved the final version for submission.

### Funding

This study was not funded.

### Data availability

The data supporting this study's findings are available from the corresponding author upon reasonable request and subject to ethical approval by the University of Energy and Natural Resources.

## Declarations

### Consent for publication

The article does not contain any individual details and consent for publication is not applicable.

### Competing interests

The authors declare no competing interests.

### Institutional review board statement

This study was conducted according to the guidelines of the Declaration of Helsinki. Ethical approval for this study was given by the Committee for Human Research and Ethics of the University of Energy and Natural Resources (CHRE/AP/288/024). Approval was also given by the research unit of the Bono Regional Teaching Hospital.

### Informed consent

Informed consent was obtained from all subjects involved in the study.

Received: 2 March 2025 / Accepted: 25 June 2025
Published online: 05 July 2025

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